

Assignment of
Artificial Intelligence

Skill Advisor AI Agent

(Rule Based Ai Agent)

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5th Semester Sec: B

1. Problem Identification in Terms of an Agent

There are countless skills a person could learn, but not every skill fits their current interests, career goals, existing skill level, or the time they actually have. Figuring this out alone usually means a lot of guesswork and research. This is a good fit for an intelligent agent, since it just needs to look at a learner's profile and a possible skill, weigh them against a set of rules, and decide whether that skill is worth pursuing right now.

The agent built here **Skill Advisor Agent** is a rule-based agent. It doesn't learn from data or improve over time instead it works off a fixed set of IF-THEN rules covering interest, career relevance, skill level, and time availability. Given a learner's profile and a skill they are considering it checks these rules and returns a recommendation along with a short reason for why it is best for them.

2. Problem Formulation

Input: The learner's current skill level, interest area, career goal, time available, and the skill they are considering.

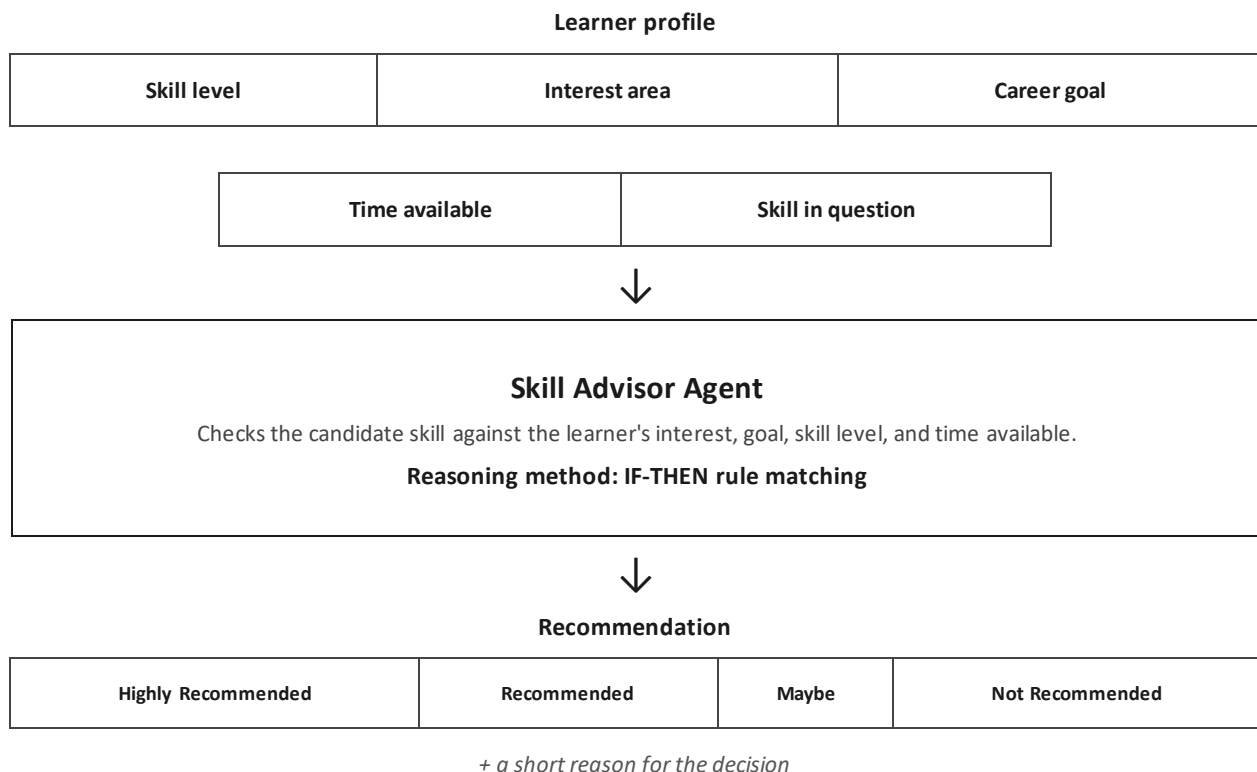
Dataset: A set of rules connecting interest, career goal, skill level, and time availability to a recommendation. Example:

Skill Level	Interest Area	Career Goal	Time Available	Candidate Skill	Rule / Condition	Recommendation
Beginner	Programming	Data Scientist	High	Python Basics	Interest and goal both match, beginner-friendly	Highly Recommended
Intermediate	Design	Marketing	Low	Advanced Photoshop	Goal does not match, time insufficient	Not Recommended
Beginner	Public Speaking	Sales	Medium	Public Speaking Basics	Interest and goal both match	Highly Recommended
Advanced	Programming	Data Scientist	Medium	HTML Basics	Skill level already exceeds candidate skill	Not Recommended
Beginner	Finance	Data Analyst	High	Excel Basics	Goal relevant, beginner-friendly	Recommended

Algorithm: A rule-based approach using IF-THEN matching the learners input is checked against the rule set to reach a decision.

Output: One of four labels: Highly Recommended, Recommended, Maybe, or Not Recommended along with a short reason.

Graphical Representation



3. Evaluation in the Context of PEAS

PEAS stands for Performance measure, Environment, Actuators, and Sensors. A standard way of describing what an agent works with and how it acts.

PEAS Element	Description for the Skill Advisor Agent
Performance Measure	How well the recommendation fits the learner's interests and goals, whether the skill level makes sense, and how quickly the answer comes back.
Environment	The learner's profile (skill level, interests, goals, time available), the skills database, and the interface used to enter details and view results.
Actuators	The screen that displays the recommendation label (Highly Recommended / Recommended / Maybe / Not Recommended) along with a short reason.
Sensors	The input form that collects the learner's skill level, interest area, career goal, time availability, and the skill they are asking about.

4. Environment Properties

The environment can also be described using the standard characteristics used for intelligent agents explained below.

Observability: (Partially observable)

The agent only knows what the learner tells it. It cannot sense things like real motivation or skills the learner practiced elsewhere but never mentioned so its picture of the learner is incomplete.

Determinism: (Deterministic within the rule base)

Since its rule-based, the same profile and the same skill will always trigger the same rules and give the same recommendation. There is no randomness in how it decides.

Episodic vs. Sequential: (Episodic)

Each skill check is its own episode. What it recommended for one skill has no effect on the next query, even though the learners saved profile carries over between them.

Static vs. Dynamic: [Semi-static (mostly static)]

The learner's profile and the skills data stay fixed while a single recommendation is being worked on. Things only change between sessions, for example if the learner updates their goal or picks up a new skill.

Discrete vs. Continuous: (Discrete)

Skill levels, interest areas, time availability, and recommendation labels are all fixed categories rather than continuous values, and the rules are written around these categories.

Known vs. Unknown: (Known)

The rules linking interest, goal, skill level, and time to a recommendation were decided in advance by the designer so the agent already knows how to respond before it runs.

Single-agent vs. Multi-agent: (Single-agent)

It acts as a personal advisor for one learner at a time and has no need to cooperate or compete with any other agent while making a decision.